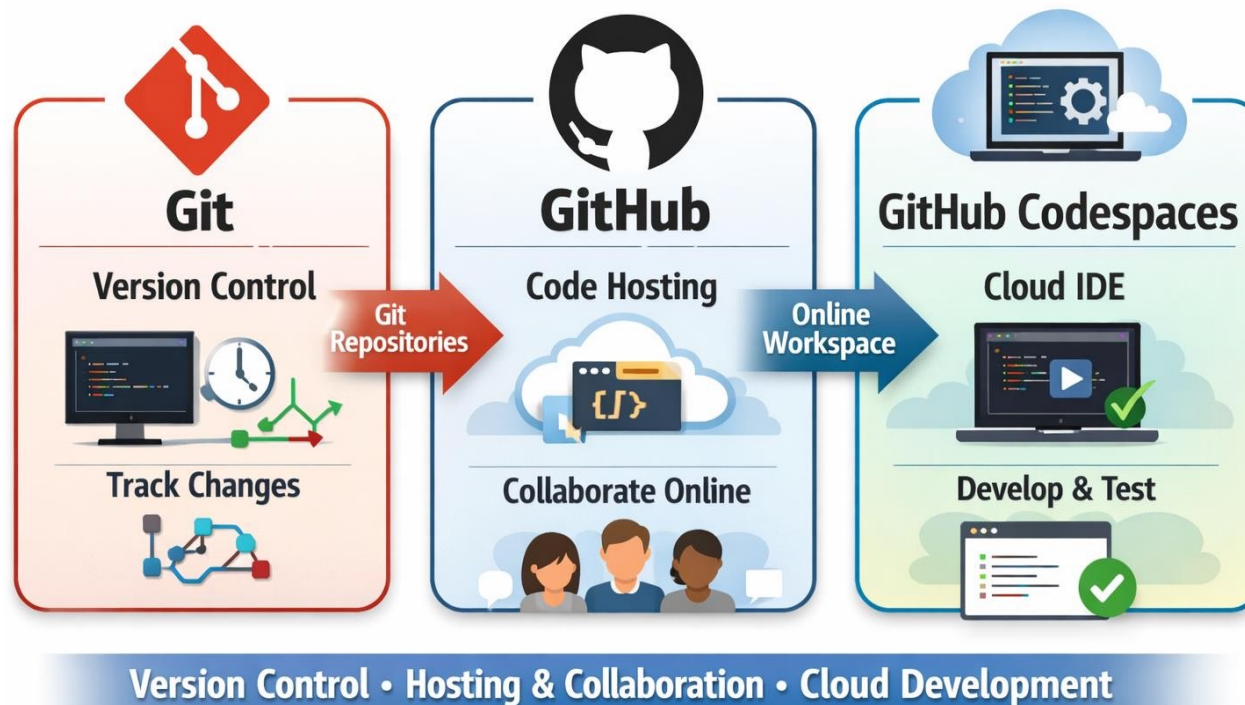


Git, GitHub and CodeSpaces



What is Git?

- Git is a **free**, open-source distributed **version control system** created by Linus Torvalds in 2005.
- It **tracks changes** in source code and enables **efficient collaboration** among multiple developers.
- Unlike centralized systems (e.g., SVN), Git stores the full project history locally, allowing offline work and full access to version history.



What is Version Control?

- The management of changes to documents, computer programs, large websites, and other collections of information



Terms

- Directory – Folder
- cd – change directory
- Terminal or command line – Interface for text commands
- CLI – Command Line Interface
- Repository – Project, or the folder where your project is kept
- Github – a website for your remote git repository

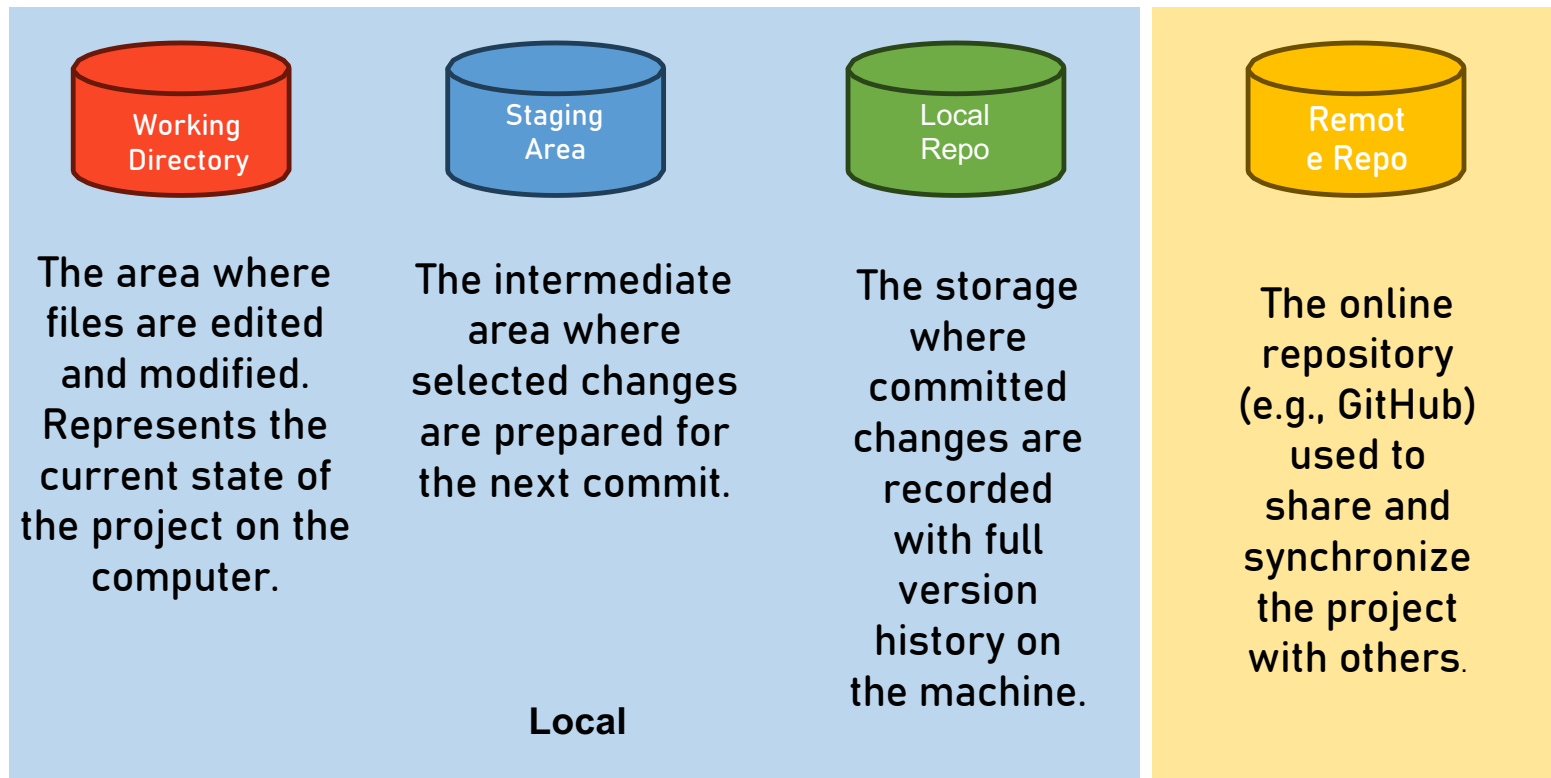
Git Commands

- Clone – Bring a repository that is hosted somewhere like Github into a folder on your local machine
- Add – Track your files and changes in Git
- Commit – save your files in Git
- Push – Upload Git commits to a remote repository, like Github
- Pull – Download changes form remote repo to your local machine (the opposite of push)

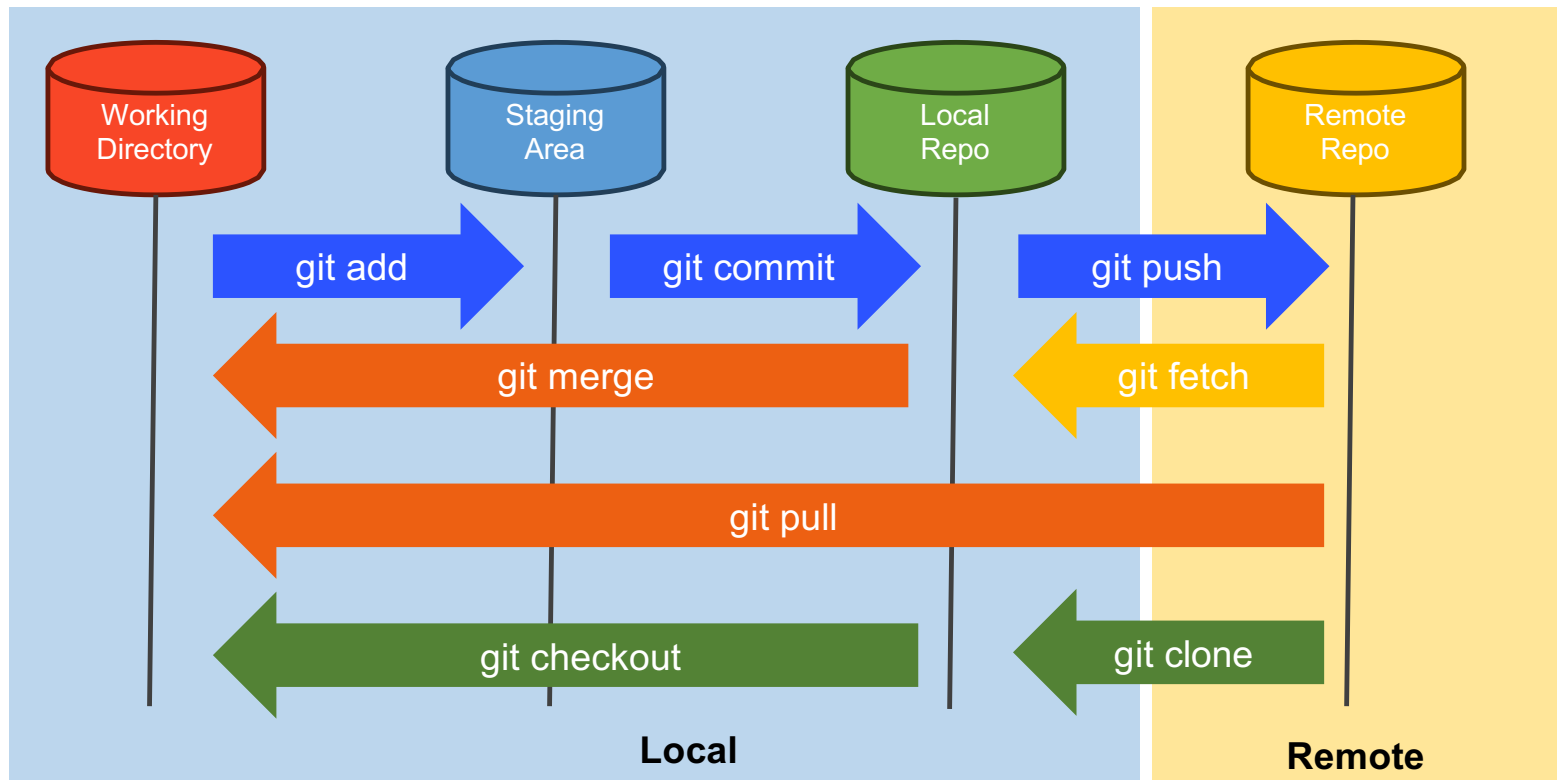
Git Commands

- Branche and merging – breaking and reconnection to the main branch
- Fork – A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project
- init – use this command inside of your project to turn it into a Git repository and start using Git with that codebase

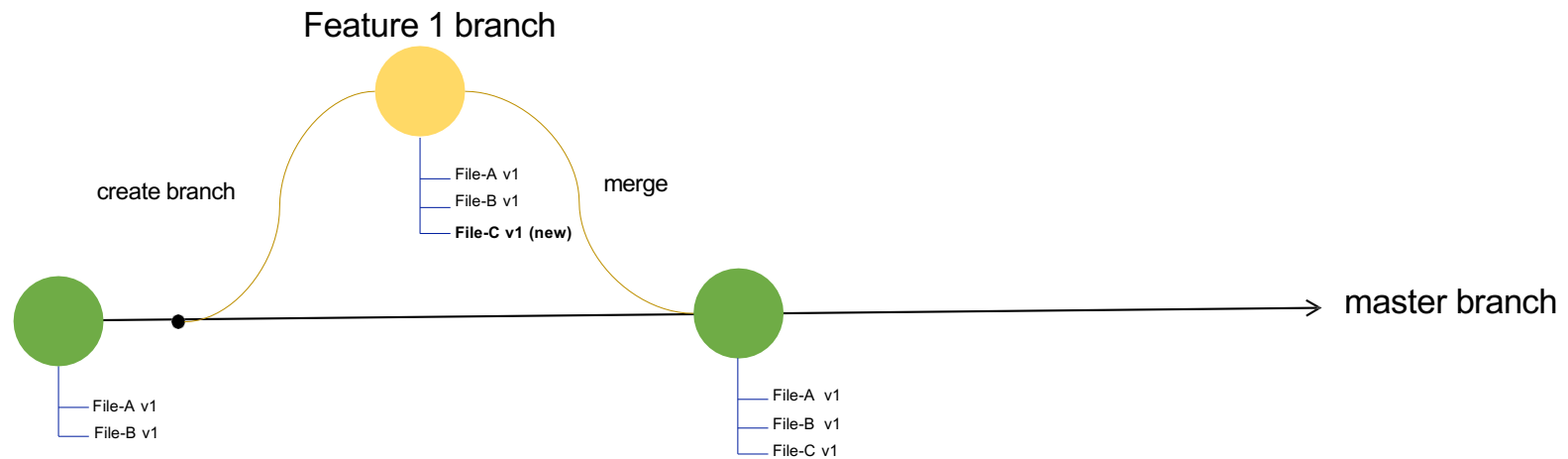
How Git works?



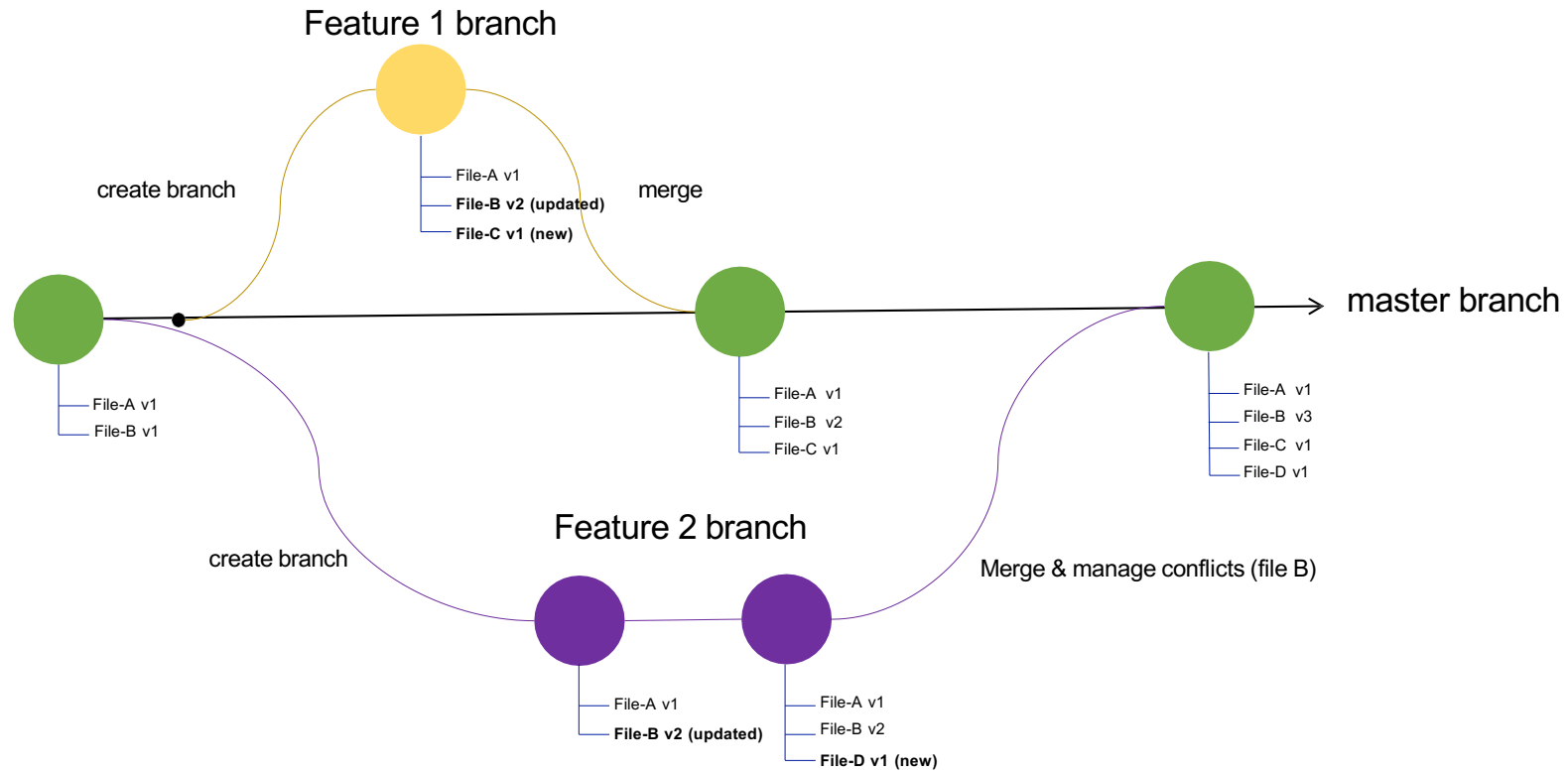
How Git works?



How git works

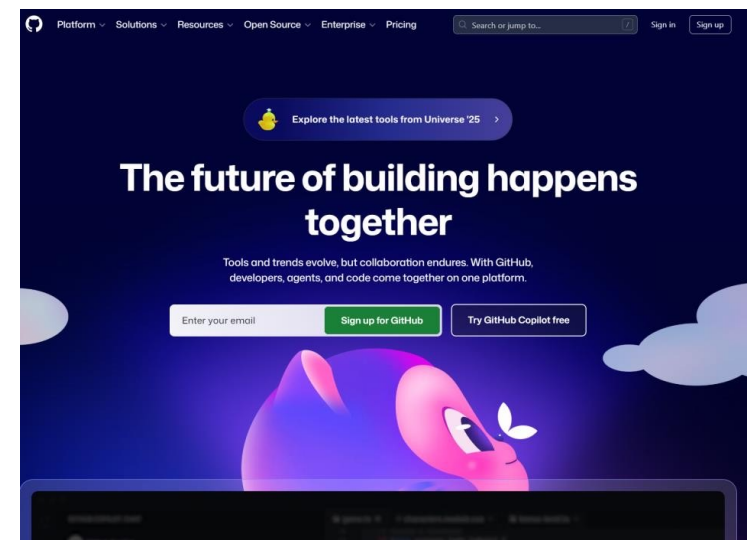


How git works

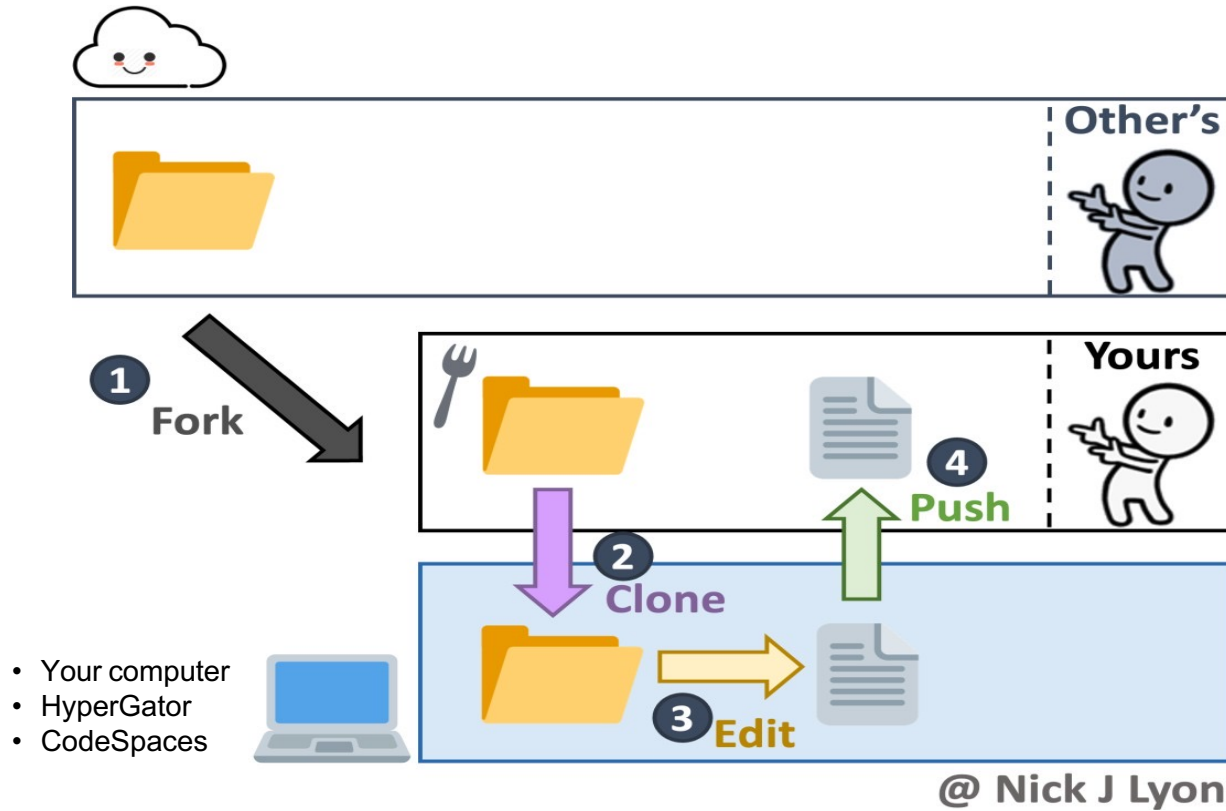


GitHub

- GitHub is an **online platform** for storing and managing projects and code.
- It allows you to **share** your work and **explore** projects created by others.
- It supports **collaboration**, enabling multiple people to work on the same project safely.
- It is widely used by students, researchers, and companies to organize and develop projects.



fork



HANDS ON!



Source: freepik

THANK YOU

Questions



Thank You!



Connect with Us!



We admonish you to live
for Jesus and connect
with our community!



 New Life Research Foundation (NELIREF) Ltd/Gte

 contact@neliref.org |
newliferesearchfoundation2013@gmail.com

 www.neliref.org

